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Vacuum hose adaptor

Abstract

A vacuum hose adaptor (**20**) having a suction air channel (**24**) defined between a sealing plate (**25**) and a male hose coupler (**22**). The sealing plate (**25**) is designed to make a relatively air tight seal against the bottom plate (**58**) (i.e., rotary brush opening, suction inlet) on the bottom of a main nozzle (**55**) on an upright vacuum cleaner (**50**) and may use one or more locking mechanisms (**26**) for temporarily holding the sealing plate (**25**) against the bottom plate (**58**) of the main nozzle (**55**). A vacuum hose (**60**) can be removably attached to the male hose coupler (**22**) so that suction air produced by the upright vacuum cleaner (**50**) can be conducted from the upright vacuum cleaner (**50**) through the main nozzle (**55**), through the suction air channel (**24**), through the male hose coupler (**22**) and through the vacuum hose (**60**) for use.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This Application claims priority from U.S. Provisional Application 63/437,643 filed on Jan. 7, 2023 and is hereby incorporated by reference in its entirety, including any figures, tables, equations or drawings.

BACKGROUND
(1) The present disclosure is related to vacuum cleaner accessories, and more specifically vacuum accessories that attach over the ‘roller brush’ opening of an upright vacuum cleaner to allow attachment of a vacuum hose to the vacuum cleaner.
(2) Since the dawn of time, man has strived to invent things to improve his surroundings and make life better. The broom was invented over 4,000 years ago to help make a residence cleaner and more appealing to live in. In the 19.sup.th century, mechanical rolling sweeping devices were invented to whisk debris directly into a containment tray. Then the final evolution was when suction was added first by hand bellows, then finally gas powered motor and then evolved with smaller portable electric motors initially called ‘Electric Suction Sweepers’, then later, the term ‘Vacuum Cleaner’. Once a tool for the affluent, it steadily became more affordable for the average household post-World War II. Over the years vacuums have become easier to use and many labor saving attachments were invented enabling the user to clean and maintain a variety of both hard and non-hard surfaces. Certain brands of known upright vacuum cleaners, for example the Kirby brand, have attachments that offer great versatility for the user. However, conversion from an

upright carpet/rug mode to hose functions has been noticeably less convenient, with the hose conversion typically requiring no less than 10 steps to transition to full hose operation and back again to the upright carpet/rug vacuuming mode (upright mode).
(3) The vacuum hose adaptor presented herein greatly simplifies the conversion process with only two steps to convert to hose operation and two more steps to return to upright operation. This saves a noticeable amount of time and labor because many of the steps removed from the current configuration are labor intensive. The labor intensive steps eliminated include: 1) twisting the belt lifter to pull the belt away from the main nozzle, 2) releasing the clamp on the main nozzle, 3) lifting the main nozzle away from the vacuum and placing it out of the way, 4) connecting the hose to the vacuum and 5) then locking it in place with the clamp. To return to upright mode, these steps can be done in reverse. The presently disclosed vacuum hose adaptor requires only two steps: 1) slipping the hose adaptor into place, and 2) flipping the actuator tab to lock the vacuum hose adaptor in place. To return to upright mode these two steps can be done in reverse. Besides decreasing the work needed for converting back and forth between modes, the disclosed vacuum hose adaptor also provides a much more stable base for rolling. This allows the user to pull the vacuum hose from nearly any direction while reducing the danger that the main vacuum will accidently fall over.
SUMMARY
(4) The disclosed vacuum hose adaptor allows a more rapid and less labor intensive transition between an upright vacuuming mode and a hose cleaning mode of an upright vacuum cleaner. The disclosed vacuum hose adaptor can comprise a housing body, a pair of swivel wheels mounted to the housing body, a locking mechanism, a pivotal arm assembly and a male hose coupler. Housing body can comprise a suction air passageway that extends between a sealing surface defined on the housing body and the male hose coupler. The sealing surface can include a rubber or foam gasket that is designed to seal against the suction inlet on a vacuum cleaner agitator (i.e., bottom plate). This transfers the suction air to the male coupler which is adapted for attachment of a vacuum hose. The swivel wheels are mounted to the bottom of the housing of the connector at an angle so that their pivotal axis of the swivel wheels are substantially perpendicular to the floor surface when mounted to the upright vacuum (e.g., an upright vacuums may be angled back at about 15 degrees due to the height of the disclosed vacuum hose adaptor mounted beneath it). The wide spacing of the pair of swivel wheels provide greater stability for the vacuum when being pulled around by the attached vacuum hose. The vacuum hose can remain attached to the disclosed vacuum hose adaptor when removed from the upright vacuum (i.e., in upright/carpet mode) to saving time when switching between modes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.
(2) FIG. **1**A are a bottom/front perspective schematic illustration of the disclosed vacuum hose adaptor. The pivotal arm assembly in a stowed position.
(3) FIG. **1**B are a top/front perspective schematic illustration of the disclosed vacuum hose adaptor. The pivotal arm assembly in a stowed position.
(4) FIG. **2** is a perspective top/rear schematic illustration of the disclosed vacuum hose adaptor showing its pivotal arm assembly in its operational position with solid lines, and one of its stowed positions with shadow lines.

(5) FIG. **3** is a left-side schematic illustration of the disclosed vacuum hose adaptor with its pivotal arm in its operational position.
(6) FIG. **4** is a perspective schematic illustration of the disclosed vacuum hose adaptor installed on an upright vacuum with an attached vacuum hose.
(7) TABLE-US-00001 DRAWING REFERENCE NUMBERS 20 Vacuum Hose Adaptor 21a Securing Indent (High-speed operation) 21 Adaptor body 21z Securing Indent (stowage) 21y Securing Indent (Low-speed operation) 23 Air passageway (front surface) 22 Male hose coupler 24a-b Pivot holes for Pivot arm 32 24 Suction air channel 25a Pivotal support (locking mechanism 26) 25 Sealing plate (seals against the roller 26 Locking mechanism brush suction inlet at the bottom plate 58) 26b Locking pin for Locking mechanism 25b Sealing gasket 27b Right exterior guide 26a Actuator Tab (locking mechanism) 28b Right attachment bracket 27a Left exterior guide 29b Right internal guide 28a Left attachment bracket 29a Left internal guide 30 Pivotal arm assembly 32 Pivotal arm 32a Pivotal axis section of arm 32 (Horiz.) 32b Contact section of arm 32 (vertical) 32c Extended section of arm 32 (Horizontal) 32y Low-speed position 32y Stowed position 33 Soft tip of arm 32 34a Back stop (back C-ring) 34b Front stop (front C-ring) 36 Friction block 40a Left swivel wheel assembly 40b Left swivel wheel assembly 42 Wheels 44 Swivel bearing 46 Swivel plate (angled) 49 Lean angle 50 Upright vacuum cleaner 51 Main Body (vacuum 50) 52 Headlight cap assembly (in up position) 53 High-speed switch (high/low switch) 54 Clamp 55 Main nozzle 57 Belt lifter 58 Bottom plate (Main nozzle 56) 60 Vacuum hose (i.e. for upright vacuum) 62a Activator arm (i.e. for button 53) 62 Female hose coupling 64 Vacuum hose conduit (hose section of 60)

DETAILED DESCRIPTION

(8) FIGS. **1**A, and **1**B are perspective schematic illustrations of a vacuum hose adaptor **20** (hereafter "hose adaptor **20**"). The hose adaptor **20** can comprise an adaptor body **21**, a male hose coupler **22**, a locking mechanism **26**, a pivotal arm assembly **30** and a pair of identical swivel wheel assemblies **40**_a_ and **40**_b_ (hereafter "swivel wheel assemblies **40**_a_ and **40**_b_"). In FIGS. **1**A and **1**B, the hose adaptor **20** is illustrated in a bottom perspective view and a top perspective view respectively. In FIGS. **1**A, the pivotal arm assembly **30** is illustrated in a secondary stowed position where the pivotal switch arm **32** (hereafter "pivotal arm **32**") can be pivoted against the right side of the adaptor body **21**. The shape of the pivotal arm assembly **30** can also allow stowage of the pivotal arm **32** in a primary stowed position **32**_z_ (hereafter "stowed position **32**_z_") to the left (see FIG. **2**) by engaging a stowage indent **21**_z_ (hereafter "indent **21**_z_"). The high-speed operational indent **21**_a_ (hereafter "indent **21**_a_") is designed to temporarily secure the pivotal arm **32** in its high-speed operational position as seen in FIGS. **2**, **3** and **4**. The low-speed operational indent **21**_y_ (hereafter "indent **21**_y_") is designed to temporarily secure the pivotal arm **32** in a low-speed operational position **32**_y_ (hereafter "low-speed position **32**_y_") as seen in FIG. **2**. The stowage indent **21**_z_ is designed to temporarily secure the pivotal arm **32** in the stowed position **32**_z_ as seen in FIG. **2**. In FIG. **1**B the pivotal arm **32** is illustrated at a position between the low-speed operational position **32**_y_ and the stowed position **32**_z_. The two stowed positions are made possible due to the shape of pivotal arm **32** the shape of the adaptor body **21**. The low-speed position **32**_y_ places the pivotal arm in free space during operation (i.e., not pressing a high-speed button) so that the vacuum cleaner to which it is attached vacuum operates at a lower speed (i.e., less suction).
(9) In FIGS. **1**A and **1**B, the male hose coupler **22** is fluidically connected to the adaptor body by a short conduit **23** (hereafter "conduit **23**"). The conduit **23** is in fluid communication with a suction air channel **24** within the adaptor body **21** for conducting suction air to the male hose coupler **22** and ultimately to a vacuum cleaner hose **60** (see FIG. **4**). The male hose coupler **22** and vacuum cleaner hose **60** (hereafter "vacuum hose **60**") can use various prior art connectors and couplers for attaching these two components together (e.g., male and female couplers **22** and **62**).
(10) The adaptor body **21** can comprise the indents **21**_a_, **21**_y_, and **21**_z_, the suction air channel **24**, a front and back pivot hole **24**_a_ and **24**_b_ respectively on each side of the suction air channel **24**, a

sealing plate **25**, a pivotal support **25***a*, a sealing gasket **25***b*, a left and right exterior guide **27***a* and **27***b* respectively (hereafter "exterior guides **27***a-b*"), a left and right attachment bracket **28***a* and **28***b* respectively (hereafter "attachment brackets **28***a-b*") and a left and right internal guide **29***a* and **29***b* respectively (hereafter "interior guides **29***a-b*"). The sealing gasket **25***b* can be bonded to the top surface of the sealing plate **25** and extends around the sealing plate **25** along its upper surface (i.e., sealing surface). The sealing gasket **25***b* can comprise nearly any compliant material, including but not limited to, TPR, TPU, foam rubber, rubber, various foam elastomers, various elastomers, etc. The exterior guides **27***a-b*, attachment brackets **28***a-b* and interior guides **29***a-b* can all be designed to interact with the lower portion of a vacuum cleaner main nozzle **55** to help guide the hose adaptor **20** in place for use (see FIG. **4**). The internal locking mechanism **26** can comprise an actuator tab **26***a* and a locking pin **26***b* which can extend through the pivotal support **25***a* to allow the locking pin **26***b* to engage the inside surface of the vacuum cleaner main nozzle **55** (hereafter "main nozzle **55**") of an upright vacuum cleaner **50** (hereafter "vacuum cleaner **50**"). The main nozzle **55** comprises a bottom plate **58** that can act as a guard for a rotary brush within the main nozzle. The swivel wheel assemblies **40***a* and **40***b* can each comprise a wheel **42**, a swivel bearing **44** and a swivel plate **46**. The swivel plate **46** of each swivel wheel assembly **40***a* and **40***b* is attached on the left and right portions of the underside of the adaptor body **21**.

(11) In FIGS. **1**A, **1**B and **3**, we see that the pivotal arm assembly **30** can comprise the pivotal arm **32** previously mentioned, a soft tip **33** on the end of the pivotal arm **32**, a back stop **34***a*, a front stop **34***b* and an optional friction device **36** (hereafter "friction block **36**"). The pivotal arm **32** can be made from various high-strength and impact resistant material, including but not limited to, a heavy gage stainless steel wire, a piano wire, an alloy wire, a polymer, a composite or nearly any other high-strength material. The soft tip **33** can be mounted at the exposed end of the pivotal arm **32** to prevent scratching of the vacuum's surface during mounting and dismounting of the hose adaptor **20**. The soft tip **33** can be made of nearly any soft plastic, such as but not limited to, nylon, polypropylene, polyester and various elastomers. The soft tip **33** may also comprise an elastomer, such as but not limited to, natural rubber, TPEs (thermal plastic elastomers), TPRs (thermal plastic rubbers) and TPUs (thermal plastic urethanes). The pivotal arm **32** can comprise three distinct sections, a pivotal axis section **32***a*, a contact section **32***b* and an extended section **32***c*. The pivotal axis section **32***a* can be designed to fit through the pivot holes **24***a-b* to provide a hinge support for the pivotal arm **32** to rotate about. The back stop **34***a* (hereafter "stop **34***a*") and front stop **34***b* (hereafter "stop **34***b*") can prevent the pivotal axis section **32***a* of pivotal arm **32** from sliding back and forth within the holes **24***a-b* and thus maintain the proper position of the pivotal axis section **32***a* with respect to the adaptor body **21**. With the pivotal axis section **32***a* having a stable position (i.e., with stops **34***a-b*) the contact axis section **32***b* can provide friction contact against the front surface of the adaptor body **21** and can have secure stops when engaging the indents **21***a*, **21***y* and **21***z*. This provides secure holding positions for the pivotal arm **32** that can temporarily hold the pivotal arm **32** at the proper orientation depending on the desired function. The indents **21***a*, **21***y* and **21***z* are illustrated here as notches in the front surface of the adaptor body **21**, but can comprise nearly any other temporary latching system that can temporarily position the pivotal arm **32** respectively in its high-speed operational position (see FIGS. **2**, **3** and **4**), its low-speed position **32***y* (see FIG. **2**) and/or its stowed position **32***z* (see FIGS. **1**B and **2**). The indents **21***a*, **21***y* and **21***z* are optional, but can make the vacuum hose adaptor **20** easier to install and operate by the user. The optional friction block **36** can be positioned against the pivotal axis section **32***a* to provide friction to allow the user to position the pivotal arm **32** in any location between the stowed position **32***z* and its high-speed operational position seen in FIGS. **2**, **3** and **4**). In some embodiments the friction block **36** can replace the need for the need for the indents **21***a*, **21***y* and **21***z*. In other embodiments, the indents **21***a*, **21***y* and **21***z* can be omitted and simply friction of the contact section **32***b* against the front surface of the adaptor body **21** can be sufficient to hold the pivotal arm in place during installation, use, and stowage. In some embodiments, only the high-speed operational indent **21***a*

can be used to provide the proper position of the pivotal arm **32** for installing the vacuum hose adaptor **20** on a upright vacuum (e.g., the upright vacuum cleaner **50** in FIG. **4**).

(12) As seen in FIGS. **1**A, **2** and **4**, the locking mechanism **26** can be mounted through the pivotal support **25***a* to provide a secure pivotal mounting structure for both the actuator tab **26***a* and the locking pin **26***b*. The actuator tab **26***a* is securely attached to the locking pin **26***b* so that the two rotate as one. In this way, rotation of the actuator tab **26***a* by the user causes the locking pin **26***b* to also rotate and securely lock the vacuum hose adaptor **20** temporarily to the bottom of the upright vacuum cleaner **50** (see FIG. **4**). The actuator tab **26***a* can include a rubberized cover to provide a soft surface for the user press against when locking and unlocking the locking pin **26***b* from the vacuum cleaner **50**. FIG. **4** illustrates the locking pin **26***b* (in shadow lines) locked in place behind the bottom plate **58**. In some embodiments, two or more locking mechanisms **26** can be used. For example, one locking mechanism **26** on the right of male hose coupler **22** as shown, and another similar locking mechanism **26** mounted on the left of the male hose coupler **22**. This can enhance the stability of the attachment of the hose adaptor **20** to the vacuum cleaner **50** when rolling on the floor (i.e., when being used, see FIG. **4**). Also, the exterior guides **27***a-b* and/or the interior guides **29***a-b*, if used, can further prevent the hose adaptor **20** from slipping around from forces caused by pulling on the attached vacuum hose **60** from various directions.

(13) The swivel wheel assemblies **40***a* and **40***b* can comprise prior art swivel wheels that can be mounted to the bottom of the adaptor body **21** with the swivel plate **46** at a lean angle **49**. The lean angle **49** is selected to orient the axis of rotation for the swivel bearing **44** to be substantially vertical during operation. That is, when the hose adaptor **20** is installed on the vacuum cleaner **50** the height of swivel wheel assemblies **40***a* and **40***b* cause the vacuum cleaner **50** to be angled up in the front, which causes the adaptor body **21** to also angle back by the lean angle **49**. Thus, the lean angle **49** orient the swivel plate **46** substantially horizontally during operation and can allow the swivel wheel assemblies **40***a* and **40***b* to swivel in every direction equally well (i.e., to provide optimal operation of the swivel wheel assemblies **40***a-b*). The sealing plate **25** can be locked against the bottom plate **58** (see FIG. **4**) with the locking mechanism **26** (see FIGS. **1**A-B, **2** and **4**). With the sealing plate **25** locked against the bottom plate **58**, the suction air from the vacuum cleaner **50** can be conducted to the male hose coupler **22** (see FIG. **1**A through **3**) and into any attached vacuum hose **60**. Note that FIG. **3** does not show the hose adaptor **20** in its operational orientation. The illustration in FIG. **3** needs to be rotated counter-clockwise by the lean angle **49** to place it in its operational orientation where swivel plate **46** is substantially horizontal (i.e., normal to the ground) and the rotational axis of the swivel bearing **44** is substantially vertical.

(14) In FIG. **3**, illustrates a side view of the hose adaptor **20** with the pivotal arm **32** in its high-speed operational position when attached to vacuum cleaner **50** (see FIG. **4**). Notice the lean angle **49** on the swivel wheel assembly **40***a* (swivel wheel **40***b* hidden behind swivel wheel assembly **40***a*). Not all hidden lines are illustrated in this figure to keep the drawing less cluttered and to focus on the swivel wheel assembly **40***a* and pivotal arm assembly **30**. The straight pivotal axis section **32***a* can be seen passing through pivot holes **24***a-b* in the suction air channel **24**. The stops **34***a* and **34***b* prevent the pivotal arm **32** from sliding either left or right and only allows it to pivot. The stops **34***a* and **34***b* can comprise washers, c-clips or other retaining devices. In this example, the stops **34***a-b* comprise a washer with a retaining clip secured in a groove in the pivotal axis section **32***a*. This is a common practice for many shafts that need to rotate while also maintaining its axial position against axial forces.

(15) FIG. **4** is a perspective schematic illustration of the disclosed vacuum hose adaptor **20** installed on the vacuum cleaner **50** with the vacuum hose **60** attached to the male hose coupler **22**. The vacuum cleaner **50** can comprise a main body **51**, a headlight cap assembly **52**, the main nozzle **55**, a high-speed switch **53**, a clamp **54**, a belt lifter **57** and the bottom plate **58**. The vacuum hose **60** can comprise a female hose coupler **62**, and activator arm **62***a* and a vacuum hose conduit **64** (i.e., a flexible vacuum hose). The female hose coupler **62** can be designed to removably attach

to the male hose coupler **22** so that suction air can be communicated to the vacuum hose conduit **64** (hereafter "flexible conduit **64**"). The basic vacuum cleaner design illustrated in FIG. **4**, has been made since 1991, but the disclosed hose adaptor **20** can be modified to fit other vacuum cleaners as well. The main nozzle **55** includes the belt lifter **57** and can be removed from the main body **51** by disengaging the clamp **54**. In this way, the main nozzle **55** can be removed and the vacuum hose **60** can be attached directly to the main body **51** for use. However, this takes many steps and requires lifting the relatively bulky main nozzle **55** and storing it somewhere. The vacuum cleaner **50** and attached vacuum hose **60** are illustrated in shadow lines to illustrate that it is prior art. Note that hidden lines for most of the hose adaptor **20** are not shown to prevent cluttering of the drawing. Only the hidden lines for wheels **42** are shown in this figure to reduce clutter and assist in understanding the illustration. The wheels **42** raise the female hose coupler **62** above the floor such that even if the activator arm **62***a* is installed downward on the male hose coupler **22** in this example to illustrate that the swivel wheels can raise the female hose coupler **62** far enough above the floor so that the activator arm **62***a* will not contact the floor even when connected pointing straight toward the floor as illustrated in FIG. **4**. The high-speed switch **53** allows the selection of running a vacuum motor within the vacuum cleaner **50** at a low-speed (lower suction) for cleaning carpets when not depressed, and at a high-speed (i.e., large suction) for using the vacuum hose **60** with various prior art tools when the high-speed switch **53** is depressed. With the main nozzle **55** attached to the vacuum cleaner **50**, the high-speed switch **53** is exposed to the outside and is not engaged which results in the vacuum cleaner **50** operating at the low-speed. When the hose adaptor **20** is installed, with the pivotal arm **32** in its high-speed position, as shown in FIG. **4**, the soft tip **33** on the pivotal arm **32** can press the high-speed switch **53** which increases the vacuum cleaner's **50** suction power by sending more electrical power to the vacuum cleaner **50**. This is possible because the pivotal arm **32** is shaped in such a way that it presses the high-speed switch **53** when installed on the bottom of the main nozzle **55**. Once the vacuum attachment **20** is secured to the main nozzle **55**, suction air can find a continuous passageway through the main nozzle **55**, sealing plate **25**, suction air channel **24**, air passageway **23**, male hose coupler **22**, female hose coupler **62** and flexible conduit **64**.

OPERATIONAL DESCRIPTION

(16) Before use, the hose adaptor **20** is slid under the main nozzle **55** with left and right exterior guides **27***a-b* assisting in guiding the attachment brackets **28***a-b* to the interior back side of the bottom plate **58**. The attachment brackets **28***a-b* can engage the interior back side of the main nozzle **55** so that the back portion of the sealing plate **25** is held against the bottom plate **58**. The weight of the vacuum cleaner **50** can then press the front portions of the sealing plate **25** and bottom plate **58** together which can place the locking mechanism **26** in the proper position for engaging the interior front portion of the main nozzle **55** and/or bottom plate **58**. Once attached, the sealing plate **25** lays flush against the bottom plate **58** (see FIG. **4**) of vacuum cleaner **50**, and if sealing gasket **25***b* is present it is pressed between the sealing plate **25** and bottom plate **58** to form a substantially airtight seal. The exterior guides **27***a-b* and interior guides **29***a-b* interact with the exterior and interior of the main nozzle **55** and bottom plate **58** respectively, and can guide the hose adaptor **20** into place under the bottom plate **58**. The left and right attachment brackets **28***a-b* can engage the back portion of the bottom plate **58** or main nozzle **55** to secure the back portion of the sealing plate **25** against the bottom plate **58**. Then the locking mechanism **26** can be engaged by the user, by pressing on the actuator tab **26***a* to pivot the locking pin above the bottom plate **58** internally to secure the front portion of the hose adaptor **20** to the main nozzle **55** and vacuum cleaner **50**. The vacuum hose **60** can be already connected to the hose adaptor **20** at male hose coupler **22** before the hose adaptor **20** is attached to the bottom of the main nozzle **55** (i.e., vacuum cleaner **50**). In other operational modes the vacuum hose **60** can be attached to the male hose adaptor **22** after the hose adaptor **20** has been attached to the vacuum cleaner **50**. Leaving the vacuum hose **60** attached to the hose adaptor **20** when not in use can save time both when re-

attaching the vacuum hose **60** to the vacuum cleaner **50** and also when switching back to an upright carpet cleaning mode. In some embodiments, the sealing plate **25** may also comprise a sealing gasket **25***b* on its top surface to improve the efficiency at which suction air is conducted into the male hose coupler **22** from the vacuum cleaner **50**.

(17) The attachment process can go as follows: (A) attach the vacuum hose to the male hose coupler **22** (note that the vacuum hose **60** may already be attached to the male hose coupler **22** to save this step when installing the hose adaptor **20**), (B) pivot the pivotal arm **32** into its selected position either low-speed (i.e., indent **21***y*) or high-speed (i.e., indent **21***a*), (C) lift the front end of the vacuum cleaner **50** slightly so that the main nozzle **55** is raised above the floor, (D) insert the back portion of the sealing plate **25** under the main nozzle **55** and allowing the exterior guides **27***a-b* to engage the exterior sides of the main nozzle **55** and the interior guides **29***a-b* to engage the interior edges (not seen) of the main nozzle **55** to guide the sealing plate into position, (E) continue sliding the sealing plate under the main nozzle **55** until the attachment brackets **28***a-b* slide over the back inner edge of the bottom plate **58** and lock in place, (F) press the sealing plate **25** against the bottom plate **58**, and then (G) press the actuator tab **26***a* to the left to rotate the locking pin **26***b* against the front underside of the bottom plate **58**. This process locks the hose adaptor **20** to the bottom of the main nozzle **55** and allows the vacuum hose **60** to pull the vacuum cleaner **50** around while cleaning. Other embodiments of the claimed invention may include the use of more than one actuator tab.

(18) To remove the hose adaptor **20** from the vacuum cleaner **50**, the actuator tab **26***a* can be rotated to the right which disengages the locking pin **26***b* from the underside of the bottom plate **58**. This releases the locking mechanism **26** from the bottom plate **58** of the main nozzle **55** and the hose adaptor **20** can then be easily slid out from under the bottom plate **58**. Once removed, the pivotal arm **32** can be pivoted down against the sealing plate **25** and snapped in place within optional indent **21***z*. The pivotal arm **32** can pivot because of the pivot holes **24***a-b* which interact with the straight pivotal axis section **32***a* of pivotal arm **32** and form a hinge. The vacuum hose **60** can be left attached to the hose adaptor **20** for storage until the next time the vacuum hose **60** is needed. Leaving the vacuum hose **60** attached can save time and effort in two ways. First, the step of attaching and detaching the vacuum hose **60** to and from the hose adaptor **20** is eliminated. Second, because the female hose coupler **62** makes a good gripping handle, having the female hose coupler **62** attached can make it easier to install the hose adaptor **20** under the main nozzle **55**.

(19) During operation of the vacuum hose **60** the vacuum cleaner **50** can be pulled around by pulling on the flexible conduit **64**. The two swivel wheels **42** can be spaced as wide apart as practical including being flush with the outer portion of the adaptor body **21** to provide a very stable support system for the front of the vacuum cleaner **50**. As the vacuum cleaner **50** is rolled around, the swivel wheels **42** can easily swivel to change direction and make it easy for the vacuum cleaner **50** to swivel and roll in any direction.

(20) Many components of the disclosed vacuum hose adaptor systems can be modified to optimize their operation. For example, in some embodiments the exterior guides **27***a* and **27***b* and interior guides **29***a* and **29***b* can all be omitted from the hose adaptor **20** and allow only the attachment brackets to help guide the hose adaptor into place. In other embodiments, only the exterior guides **27***a-b* can be used. In other embodiments, the exterior guides **27***a-b* can be located at a higher portion in the back (i.e., away from the male hose coupler **22**) to provide better contact with the sides of the main nozzle **55** to better guide the hose adaptor **20** into place for use. In other embodiments, a friction contact between the pivotal arm **32** and the adaptor body **21** is used without indents.

(21) Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The specific dimensions shown are to assist the readers in understanding of the vacuum hose adaptor structure, but should not be used to limit the scope of a particular vacuum hose adaptor structure. The particular embodiments

disclosed herein are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the illustrative embodiments disclosed above may be altered, combined or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While systems and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the systems and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some amount. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.
(22) As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

Claims

1. A vacuum hose adaptor for removable attachment to a main nozzle of a vacuum cleaner, comprising: an adaptor body, comprising a front, a back, a top and a bottom, wherein a suction air channel and a sealing plate are defined on the top of the adaptor body; a vacuum hose coupler attached to the front of the adaptor body and in suction communication with the suction air channel and configured to be removably attached to a flexible vacuum hose; a pair of swivel wheels that are attached to the bottom of the adaptor body at an angle with respect to the sealing plate; a locking mechanism configured to engage a main nozzle of a vacuum cleaner during use to removably secure the front of the adaptor body against the main nozzle; wherein suction air can be communicated between the main nozzle and the suction air channel during use of the vacuum cleaner, and a pivotal arm assembly comprising a pivotal arm with a tip; wherein the pivotal arm is configured to be movable between a stowed position and a high-speed position, wherein the tip is configured to engage a high-speed switch on the vacuum cleaner when the pivotal arm is in the high-speed position.
2. The vacuum hose adaptor of claim 1, wherein the adaptor body comprises a left side with a left guide and a right side with a right guide, wherein the left and right guides extend above the sealing plate and are configured to engage a left and right side of the main nozzle respectively.
3. The vacuum hose adaptor of claim 1, wherein the sealing plate comprises a gasket bonded to the sealing plate, wherein the gasket is configured on the sealing plate to form a seal between the sealing plate and the main nozzle of the vacuum cleaner.
4. The vacuum hose adaptor of claim 1, wherein the locking mechanism comprises: one or more attachment bracket at the back of the adaptor body configured to engage a back portion of the main nozzle; and one or more locking pins at a front portion of the adaptor body configured to removably secure the front portion of the adaptor body to a front portion of the main nozzle.

5. The vacuum hose adaptor of claim 1, wherein the adaptor body comprises a stowed indent that is configured to engage the pivotal arm when in the stowed position.
6. The vacuum hose adaptor of claim 1, wherein the adaptor body comprises a high-speed indent that is configured to engage the pivotal arm when in the high-speed position.
7. The vacuum hose adaptor of claim 1, wherein the pivotal arm is further configured to include a low-speed position.
8. The vacuum hose adaptor of claim 7, wherein the adaptor comprises a low-speed indent that is configured to engage the pivotal arm when in the low-speed position.
9. A vacuum hose adaptor for removable attachment to a main nozzle of a vacuum cleaner, comprising: an adaptor body, comprising a front, a back, a top, a bottom, a left guide and a right guide, wherein a suction air channel and a sealing plate are defined on the top of the adaptor body and the left and right guides extend above the sealing plate and are configured to engage the main nozzle; a vacuum hose coupler attached to the front of the adaptor body and in suction communication with the suction air channel and configured to be removably attached to a flexible vacuum hose; a pair of swivel wheels that are attached to the bottom of the adaptor body at an angle with respect to the sealing plate; a locking mechanism configured to engage a main nozzle of a vacuum cleaner during use to removably secure the front of the adaptor body against the main nozzle; wherein suction air can be communicated between the main nozzle and the suction air channel during use of the vacuum cleaner, and a pivotal arm assembly comprising a pivotal arm with a tip; wherein the pivotal arm is configured to be movable between a stowed position and a high-speed position, wherein the adaptor body comprises a stowed indent that is configured to engage the pivotal arm when in the stowed position.
10. The vacuum hose adaptor of claim 9, wherein the adaptor body comprises a high-speed operational indent that is configured to engage the pivotal arm when in the high-speed position.
11. The vacuum hose adaptor of claim 9, wherein the pivotal arm is further configured to include a low-speed position.
12. The vacuum hose adaptor of claim 11, wherein the adaptor comprises a low-speed indent that is configured to engage the pivotal arm when in the low-speed position.
13. The vacuum hose adaptor of claim 9, wherein the sealing plate comprises a gasket bonded to the sealing plate, wherein the gasket is configured on the sealing plate to form a seal between the sealing plate and the main nozzle of the vacuum cleaner.
14. The vacuum hose adaptor of claim 9, wherein the locking mechanism comprises: one or more attachment bracket at the back of the adaptor body configured to engage with a back portion of the main nozzle; and one or more locking pins at a front portion of the adaptor body configured to removably secure the front portion of the adaptor body to a front portion of the main nozzle.
15. The vacuum hose adaptor of claim 9 wherein the tip is configured to engage a high-speed switch on the vacuum cleaner when the pivotal arm is in the high-speed position.
16. The vacuum hose adaptor of claim 9 further comprising a friction block configured to allow the pivotal arm to be positioned in any location between the stowed position and the high-speed position.